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SIMATS ENGINEERING

CO4 - ASSESSMENT TOOL 3

Design and Develop Model

COURSE NAME: Deep Learning
SLOT :

COURSE CODE: MLA04

COURSE OUTCOME COVERED

CO 4: Students will be able to understand and apply probabilistic graphical models including Bayesian Networks, Markov Random Fields, Hidden Markov Models, and Conditional Random Fields. They will learn how to represent complex probabilistic relationships among variables and perform inference and learning in graphical models. Students will be able to use these models for reasoning under uncertainty and solving real-world decision-making problems. BL-6

Course Outcome Weightage: 10%

Duration: 90 minutes

Assessment Tool Weightage: 30%

Mark : 25

Code of Conduct:

I SHAM S. (192325076) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Name of the student:

Criteria	Article Selection & Problem Understanding (2)	Deep Learning Model & Methodology Analysis (2.5)	Results & Performance Evaluation (2.5)	Critical Analysis & Technical Evaluation (2)	Future Scope, Presentation & Documentation (1)	Total (10)
Weightage	20 %	25%	25%	20 %	10 %	100 %
Q1						
Total						

Select and review a recent research article related to the below **topic**, such as **Transfer Learning, DenseNet, PixelNet, RNN, Bidirectional RNN, Encoder–Decoder/Seq2Seq, BPTT, or LSTM**. Summarize the **problem addressed, dataset used, proposed deep learning model, methodology, performance metrics, major results, and key findings**, and critically comment on the **advantages, limitations, and possible future improvements** of the approach presented in the article.

Prepare a structured Article Review addressing the following sections:

1. Article Details

Provide the following information:

Title of the research article

Names of the authors

Journal/Conference name

Year of publication

DOI or publication link

Topic/Deep Learning technique addressed

2. Research Problem

What problem does the article attempt to solve?

Why is the problem important?

What are the limitations of existing approaches identified by the authors?

3. Objectives of the Study

What are the main objectives of the research?

What specific goals does the proposed model attempt to achieve?

4. Dataset Description What dataset(s) are used?

What is the size of the dataset?

What type of data is used—images, text, time-series, video, etc.?

How is the dataset divided into training, validation, and testing sets?

5. Proposed Methodology

Explain the overall methodology adopted in the article. Identify the deep learning architecture used.

Explain the important layers/components of the model.

Describe the data preprocessing techniques used.

Explain the training procedure and important parameters, wherever available.

6. Deep Learning Technique Used

Explain how the selected technique is applied in the research.

How is Transfer Learning used?

How does DenseNet perform feature reuse?

How is RNN/LSTM used to process sequential data?

How does a Bidirectional RNN utilize information from both directions?

How does an Encoder–Decoder/Seq2Seq architecture process input and output sequences?

How is BPTT used for training recurrent networks?

7. Model Architecture

Draw or reproduce a simplified architecture diagram of the proposed model and explain:

Input → Preprocessing → Feature Extraction/Encoder → Hidden Layers →
Classification/Decoder → Output
Identify the role of the major components in the architecture.

8. Performance Evaluation

Identify the evaluation metrics used in the article, such as:
Accuracy, Precision, Recall, F1-Score , Loss , Mean Squared Error (MSE)
BLEU Score, where applicable
Report the major performance results obtained by the proposed model.

9. Results and Discussion

What are the major findings of the research?
How does the proposed model perform compared with existing methods?
Which model/technique produced the best results?
What factors contributed to the observed performance?

10. Advantages of the Proposed Model

Identify at least three advantages of the proposed approach.

11. Limitations of the Study

Identify the major limitations mentioned by the authors and any additional limitations you observe, such as: Small dataset, High computational requirements, Training time, Overfitting
Limited generalization, Model complexity.

12. Critical Analysis

Provide your own analysis of the research article:
Is the selected deep learning architecture appropriate for the problem?
Is the dataset adequate?
Are the evaluation metrics appropriate?
Are the experimental results convincing?
Could another Unit IV technique provide better performance?

13. Future Scope

Suggest possible improvements or future research directions based on the limitations identified in the article.

14. Conclusion

Provide a brief conclusion summarizing:
The research problem
Proposed model
Major results
Your overall assessment of the article

15. References

Provide the complete citation of the selected research article and any additional references used in the review.

ARTICLE REVIEW

Selected Topic: Transfer Learning

Research Article

Title: *Enhancing Skin Cancer Classification Using Efficient Net B0-B7 through Convolutional Neural Networks and Transfer Learning with Patient-Specific Data*

Authors: Kanchana K, Kavitha S, Anoop K J, and Chinthamani B

Journal: *Asian Pacific Journal of Cancer Prevention*

Year: 2024

Volume/Issue: Volume 25, Issue 5, pp. 1795–1802

DOI: 10.31557/APJCP.2024.25.5.1795

Deep Learning Technique: Transfer Learning using EfficientNet B0–B7

The article was published in May 2024 and evaluates multiple EfficientNet variants using ImageNet pre-trained weights for skin-cancer classification.

1. Article Details

The selected research article investigates the use of **transfer learning with EfficientNet B0–B7** for automated skin cancer classification.

The researchers use pre-trained CNN models and adapt them to dermoscopic skin-lesion images. Instead of training a deep neural network completely from scratch, the knowledge learned from ImageNet is transferred to the medical-image classification task.

The study focuses on improving classification performance through:

- Image preprocessing
- Noise/artifact removal
- Data augmentation
- Transfer learning
- Fine-tuning
- Comparison of EfficientNet B0–B7

The authors evaluate the models using several performance measures, including accuracy, precision, recall, F1-score, specificity, ROC-AUC, and confusion matrices.

2. Research Problem

Problem Addressed

Skin cancer diagnosis is difficult because different skin diseases can have very similar visual appearances. There can also be significant variation within the same disease category.

Traditional visual examination depends heavily on the experience of dermatologists.

The research therefore attempts to develop an automated deep-learning approach capable of classifying different skin-cancer/skin-disease categories from dermoscopic images.

Why is the Problem Important?

Early identification of potentially dangerous skin lesions can support timely medical examination and treatment.

An automated system can potentially:

- Assist dermatologists
- Reduce manual examination effort
- Process large numbers of dermoscopic images
- Provide computer-aided diagnostic support
- Improve consistency in image classification

Limitations of Existing Approaches

The article identifies challenges including:

- Inter-class similarity between skin diseases
- Intra-class variation
- Limited availability of labelled medical images
- Class imbalance
- Image noise and artifacts
- Difficulty in distinguishing visually similar disease categories

The authors therefore investigate transfer learning and preprocessing to improve classification performance.

3. Objectives of the Study

The main objectives are:

1. To develop a deep-learning approach for skin-disease classification.
2. To investigate EfficientNet B0–B7 architectures.
3. To use **ImageNet pre-trained weights** through transfer learning.
4. To apply image preprocessing to remove unwanted artifacts.
5. To use image augmentation to increase data variation.
6. To fine-tune CNN models for dermoscopic images.
7. To compare the performance of EfficientNet variants.
8. To analyze class-wise classification performance.
9. To identify the EfficientNet model providing the best results.
10. To investigate the potential of transfer learning for medical image classification.

4. Dataset Description

The study uses dermoscopic images associated with the **ISIC/ISBI skin-lesion datasets**. The article describes the use of publicly available dermoscopic images and specifically evaluates the approach using the **ISBI-2016 test dataset**, which contains approximately **379 de-identified images** with benign and malignant skin lesions.

Type of Data

The dataset consists of:

- RGB dermoscopic images
- Skin-lesion images
- Ground-truth annotations/labels

Disease Categories

The study discusses classification involving categories such as:

- Melanoma
- Acne
- Psoriasis
- Eczema
- Keratosis
- Lichen planus
- Rosacea

The article also reports class-specific performance, showing that some classes are considerably more difficult to classify than others.

Dataset Challenge

One major challenge is **class imbalance**. For example, melanoma represents only about **2.42%** of the images in the reported analysis.

Training / Validation / Testing

The article describes training using augmented dermoscopic images and evaluation using the ISBI-2016 test dataset. A specific conventional percentage split such as 70/15/15 is **not clearly specified in the source article**, so it should not be claimed in the assignment.

5. Proposed Methodology

The overall methodology can be represented as:

Dermoscopic Images



Image Acquisition



Image Preprocessing



Noise / Hair Removal



Blurring & Thresholding



Image Augmentation



Resize / Scaling



Pre-trained EfficientNet B0–B7



Transfer Learning



Fine-Tuning



Feature Extraction



Classification



Performance Evaluation

The article applies preprocessing techniques to reduce unwanted elements such as hair and background noise. It also uses transformations such as rotation, flipping, and zooming for augmentation.

6. Deep Learning Technique Used – Transfer Learning

What is Transfer Learning?

Transfer learning is a deep-learning technique in which a model trained on a large dataset is reused for another related task.

Instead of learning all features from the beginning, the model starts with knowledge acquired from the original dataset.

In this research:

ImageNet



Pre-trained EfficientNet



Transfer Learning



Dermoscopic Images



Fine-Tuning



Skin Cancer Classification

How Transfer Learning is Applied

The researchers use **ImageNet pre-trained weights** for EfficientNet models.

The pre-trained network already contains useful visual representations.

These learned features are adapted to dermoscopic images through fine-tuning.

Benefits

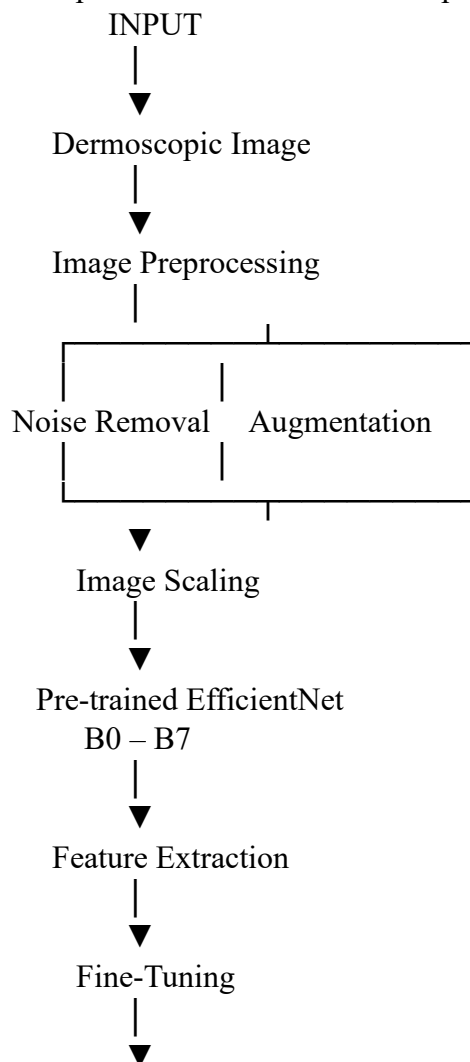
Transfer learning:

- Reduces training requirements
- Helps when medical datasets are limited
- Provides useful initial feature representations
- Can improve convergence
- Reduces the need to train a complete CNN from scratch

The article explicitly reports that transfer learning and fine-tuning improve the ability of EfficientNet models to distinguish different skin-cancer categories.

7. Model Architecture

The simplified architecture can be represented as:



Classification



Skin Disease Class

Major Components

Input

Receives the dermoscopic skin-lesion image.

Preprocessing

Reduces noise and unwanted artifacts.

Augmentation

Creates additional variations of existing images.

EfficientNet

Extracts visual features from the images.

Transfer Learning

Uses knowledge obtained from ImageNet.

Fine-Tuning

Adapts the pre-trained CNN to the target medical-image problem.

Classification Layer

Produces the final disease-category prediction.

8. Performance Evaluation

The article uses several evaluation metrics:

Accuracy

Measures the overall proportion of correctly classified samples.

Precision

Measures how many samples predicted as a particular class are actually members of that class.

Recall

Measures how many actual samples of a class are correctly detected.

F1-Score

Provides a balance between precision and recall.

Specificity

Measures the ability to correctly identify negative cases.

ROC-AUC

Measures the model's ability to distinguish between classes across classification thresholds.

Confusion Matrix

Shows the class-wise prediction performance and helps identify which diseases are being confused.

These metrics are particularly important because accuracy alone may not adequately describe performance when the dataset is imbalanced.

9. Results and Discussion

The study compares **eight EfficientNet models: B0, B1, B2, B3, B4, B5, B6, and B7.**

The reported top-1 and top-5 results are:

Model	Top-1 Accuracy	Top-5 Accuracy
EfficientNet-B0	81.0%	87.5%
EfficientNet-B1	82.5%	89.5%
EfficientNet-B2	82.7%	91.3%

Model	Top-1 Accuracy	Top-5 Accuracy
EfficientNet-B3	83.4%	93.2%
EfficientNet-B4	83.7%	94.5%
EfficientNet-B5	83.9%	95.3%
EfficientNet-B6	84.1%	96.7%
EfficientNet-B7	84.4%	97.1%

These results show an improvement as the EfficientNet model scale increases in the reported experiments. EfficientNet-B7 achieved the best reported top-1 accuracy of **84.4%** and top-5 accuracy of **97.1%**.

Class-wise Findings

The study reports that:

- Benign keratosis achieved accuracy above 87%.
- Eczema was more difficult to classify.
- Melanoma represented only 2.42% of images.
- Melanoma achieved an average accuracy of approximately 80.51% across the models.
- Some classes remained challenging despite the preprocessing and transfer-learning approach.

Best Model

EfficientNet-B7 produced the strongest overall reported results.

10. Advantages of the Proposed Model

1. Transfer Learning

The model can reuse knowledge learned from ImageNet rather than learning all visual features from scratch.

2. Multiple EfficientNet Variants

The study systematically compares B0 through B7, allowing the effect of model scaling to be analyzed.

3. Data Augmentation

Rotation, flipping, and zooming increase variation in training images and can improve generalization.

4. Artifact Removal

Removing hair and unwanted noise helps the CNN focus on lesion-related visual features.

5. Detailed Evaluation

The research uses multiple metrics and confusion matrices rather than relying only on accuracy.

6. Computational Efficiency

The article reports that EfficientNet-B7 achieves strong performance while being substantially smaller than the leading CNN referred to in the study.

11. Limitations of the Study

The study has several limitations.

1. Class Imbalance

Some disease categories contain significantly fewer images than others.

Melanoma is specifically reported as a minority class.

2. Difficult Disease Categories

Some classes, particularly eczema, remain difficult to classify accurately.

3. Limited Dataset

Dermoscopic medical datasets are relatively difficult to obtain and label at large scale.

4. Generalization

Strong performance on benchmark datasets does not automatically guarantee equivalent performance on images from different hospitals, cameras, devices, or patient populations.

5. Computational Cost

Although EfficientNet is designed for efficiency, larger variants such as B6 and B7 require more computational resources than smaller variants.

6. Clinical Deployment

A research classification model should not be considered a replacement for professional medical diagnosis without extensive external validation.

12. Critical Analysis

Is the Architecture Appropriate?

Yes.

EfficientNet with transfer learning is appropriate for dermoscopic image classification because the task is image-based and the available medical data is comparatively limited.

The use of ImageNet pre-trained weights provides a strong starting point for feature extraction.

Is the Dataset Adequate?

The dataset is useful as a benchmark, but it is **not sufficient by itself for universal clinical deployment**.

A stronger system would require:

- Larger datasets
- Multi-center data
- Different imaging devices
- Diverse patient populations
- External validation datasets

Are the Metrics Appropriate?

Yes.

Using accuracy together with precision, recall, F1-score, specificity, ROC-AUC, and confusion matrices is appropriate, particularly because the dataset has class imbalance.

Are the Results Convincing?

The comparison between B0–B7 is useful because it demonstrates how model scaling affects performance.

However, the reported results should be interpreted carefully because high benchmark performance does not necessarily prove clinical reliability.

Could Another Technique Perform Better?

Yes.

Other approaches could potentially improve performance, such as:

- EfficientNetV2
- Vision Transformers
- Attention mechanisms
- Ensemble learning
- Self-supervised pre-training
- Explainable AI techniques

A particularly useful improvement would be combining transfer learning with **attention and explainability**, allowing clinicians to understand which regions influenced the prediction.

13. Future Scope

The following improvements can be considered:

1. Larger and More Diverse Datasets

Use images from multiple hospitals and different dermoscopic devices.

2. Class-Balancing Techniques

Use:

- Oversampling
- Class-weighted loss
- Focal loss
- Synthetic data generation

to improve minority-class performance.

3. Attention Mechanisms

Attention can help the model focus on important lesion regions.

4. Explainable AI

Techniques such as:

- Grad-CAM
- LIME
- SHAP

could help visualize why a model made a particular prediction.

5. EfficientNetV2

Newer EfficientNet variants could be compared with the B0–B7 family.

6. External Validation

The model should be evaluated on completely independent datasets before clinical deployment.

7. Clinical Decision Support

The model could eventually be integrated into a decision-support system that assists dermatologists rather than replacing them.

14. Conclusion

The reviewed article investigates **transfer learning using EfficientNet B0–B7 for skin-cancer classification**. The researchers use ImageNet pre-trained models, image preprocessing, augmentation, and fine-tuning to address the challenges of dermoscopic image classification.

Among the evaluated architectures, **EfficientNet-B7 achieved the best reported performance, with 84.4% top-1 accuracy and 97.1% top-5 accuracy**.

Overall, the article demonstrates that transfer learning is a practical approach for medical image classification when labelled training data is limited. However, class imbalance, difficult disease categories, dataset diversity, and external clinical validation remain important challenges.

My overall assessment: The research provides a relevant and technically appropriate application of transfer learning. Its systematic comparison of EfficientNet variants is valuable, but future work should focus on larger and more diverse datasets, external validation, explainability, and clinical testing.

15. References

Primary Research Article

Kanchana, K., Kavitha, S., Anoop, K. J., & Chinthamani, B. (2024). **Enhancing Skin Cancer Classification Using Efficient Net B0-B7 through Convolutional Neural Networks and Transfer Learning with Patient-Specific Data**. *Asian Pacific Journal of Cancer Prevention*, 25(5), 1795–1802. DOI: 10.31557/APJCP.2024.25.5.1795.

[Research article – Asian Pacific Journal of Cancer Prevention](#)

Additional Reference

The study uses the **ImageNet pre-trained weights** as the source for transfer learning and evaluates the EfficientNet family for dermoscopic image classification.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Article Selection & Problem Understanding	20%	Comprehensive understanding of real-world problem and constraints	Good understanding with minor gaps in requirements	Basic understanding; some requirements unclear	Poor understanding; misinterprets problem context
Deep Learning Model & Methodology Analysis	25%	Applies advanced and relevant concepts effectively to solve the problem	Applies appropriate concepts with minor errors	Limited application of concepts	Incorrect or no application of concepts
Results & Performance Evaluation	25%	Innovative, practical, and feasible solution aligned with industry standards	Logical and feasible solution with minor gaps	Basic solution with limited feasibility	Unclear or impractical solution
Critical Analysis & Technical Evaluation	20%	Effective use of modern tools, technologies, and real data	Appropriate use of tools with correct implementation	Limited or inconsistent use of tools	Minimal or incorrect use of tools
Future Scope, Presentation & Documentation	10%	Thorough analysis with validated results and performance evaluation	Good analysis with reasonable validation	Basic analysis with limited validation	Poor or no analysis